

Vesham Thanujha Yadav

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GitHub: <https://github.com/Seagull28> | Location: Hyderabad, Telangana, India

EDUCATION

MVSR Engineering College | B.E. Computer Science-AIML | GPA: 7.90/10.0 | Expected: Sep 2027
Samskruti College of Engineering | Diploma in Computer Science | GPA: 9.67/10.0 | Graduation: Aug 2023

PROJECTS

Melodex : Cross-Cultural Music Transcription (In Progress) | *Python, Demucs v4, Basic Pitch, FFmpeg*

- Architecting a high-performance audio intelligence pipeline to isolate 6 neural stems (Vocals, Drums, Piano, Guitar, Other) using the Demucs v4 Hybrid Transformer.
- Implement translator logic to extract high-fidelity MIDI data.
- Engineering a segment-chunking execution engine enabling end-to-end audio processing on 4GB RAM hardware; currently benchmarking MIDI transcription accuracy across multilingual vocal styles.

Face Recognition Based Attendance System | *Django 5.2, SQLite, OpenCV, Redis*

- Engineered real-time face recognition pipeline processing frames at 3–12 FPS, achieving first confirmed attendance match in under 1.5 seconds for cohorts of up to 72 students.
- Optimized performance via quarter-resolution processing and frame-skipping, reducing per-frame CPU load to 250–600 ms on standard hardware with linear scaling to 200+ students.

College Bus Tracking System | *Kotlin, Firebase, OSMDroid*

- Developed a native Android application supporting up to 50 simultaneous buses/routes, validated through unit testing and VirtualAppTest; architecture designed to scale for larger fleets with backend tuning.
- Integrated OSMDroid for live map rendering and route visualization; bus locations update every 5–10 seconds via Firebase, delivered instantly to students through real-time listeners.

Case Study: Terrain Relative Navigation (TRN) | *NumPy, SciPy, Matplotlib*

- Validated autonomous navigation accuracy in GPS-denied environments by implementing crater-based landmark detection using NumPy and SciPy.
- Enhanced feature extraction by analyzing geometric matching algorithms for planetary surface images and visualizing alignment results via Matplotlib.

LEADERSHIP & ACTIVITIES

Street Cause - Vice President, Skill Development (Student Run NGO) | Aug 2025 – Present

- Lead a team of 8 members in executing community development initiatives, focusing on fundraising strategies and comprehensive needs assessment through survey reports.
- Conduct field surveys to identify community needs and develop data-driven proposals for sustainable development interventions.

SKILLS

- Programming Languages:** Python, Java, C, C++, Kotlin
- Frameworks & Libraries:** Django 5.2, OpenCV, face_recognition, Demucs v4, Basic Pitch
- Tools & Databases:** Android Studio, Firebase (Realtime DB + Auth), Redis, SQLite, MySQL, FFmpeg
- Web & UI:** HTML, CSS, OSMDroid
- Languages:** English, Telugu, Hindi